HEART DISEASE PREDICTION

Using Machine Learning — Comparative Analysis

Submitted By

Ashi Bansal | 23FE10CSE00394

Ridhi Kulshreshtha | 23FE10CSE00560

Course: Machine Learning Lab

B.Tech — Computer Science & Engineering

SECTION: J (6th semester) Batch-2

Academic Year 2024–25

MANIPAL UNIVERSITY JAIPUR

Submitted To

Dr. Arpita Baronia

Department of Computer Science and Engineering
Manipal University Jaipur
Jaipur, Rajasthan

1. Abstract

Cardiovascular disease is one of the leading causes of mortality worldwide, making early and accurate detection a critical clinical need. This project presents a comparative analysis of ten machine learning classification algorithms — Logistic Regression, Naive Bayes, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Decision Tree, Random Forest, XGBoost, CatBoost, AdaBoost, and Gradient Boosting — applied to a merged heart disease dataset (Statlog, Cleveland, and Hungary). The dataset underwent careful preprocessing including outlier capping, missing-value imputation, feature selection based on correlation, and a custom ColumnTransformer pipeline combining StandardScaler and OneHotEncoder. Among all models evaluated, KNN achieved the highest test accuracy of approximately 90.3%, while ensemble methods such as XGBoost and CatBoost closely followed at ~87.2%. The study demonstrates that proper preprocessing and feature engineering significantly enhance model performance for binary heart disease classification.

2. Literature Review

Several studies have explored the use of machine learning techniques for heart disease prediction. Logistic Regression has traditionally been used due to its interpretability, but its performance is limited when relationships between variables are non-linear. Decision Trees and Random Forests improve upon this by capturing hierarchical decision boundaries.

More recently, ensemble boosting methods such as XGBoost and CatBoost have demonstrated superior performance due to their ability to iteratively correct errors made by previous models. K-Nearest Neighbor (KNN), although simple, has also shown strong results in smaller datasets due to its instance-based learning approach.

Despite these advancements, there is no universally best algorithm. Model performance varies depending on dataset characteristics, preprocessing techniques, and feature selection methods. This reinforces the need for a comparative evaluation under standardized conditions.

3. Introduction

Problem Statement

Heart disease remains the single largest cause of death globally. Timely diagnosis is often hampered by the high cost of specialist consultations and the subjective nature of clinical assessment. Machine learning offers a data-driven approach to predict the likelihood of coronary heart disease (CHD) from routinely collected patient parameters, enabling faster, more objective screening decisions.

Importance of the Topic

Early prediction of heart disease can drastically reduce mortality by enabling preventive interventions. Traditional diagnostic methods require multiple invasive tests; ML-based models can generate risk scores from non-invasive parameters like chest pain type, resting ECG results, and maximum heart rate. A comparative study is especially valuable because no single algorithm

dominates across all datasets — understanding which model suits a given clinical setting empowers practitioners to make informed deployment choices.

This project uses the merged Heart Statlog Cleveland Hungary Final dataset (1,190 records, 12 features), selects the five most correlated features with the target variable, and rigorously evaluates ten classifiers under identical preprocessing conditions to identify the best-performing model.

Machine learning has emerged as a powerful tool in healthcare analytics due to its ability to identify complex patterns in large datasets. Unlike traditional statistical methods, ML models can capture non-linear relationships between clinical variables and disease outcomes. In recent years, ML-based decision support systems have gained traction for assisting physicians in diagnosis, prognosis, and treatment planning.

In the context of cardiovascular diseases, ML models can analyze patient attributes such as heart rate, ECG readings, and exercise tolerance to predict disease risk. These systems can significantly reduce diagnostic time and improve accessibility, especially in resource-constrained environments. However, the effectiveness of such models depends heavily on data preprocessing, feature selection, and algorithm choice, which motivates the comparative analysis conducted in this study.

4. Methodology / Working

Dataset

The dataset used is heart_statlog_cleveland_hungary_final.csv — a consolidated file combining three widely-used heart disease datasets, containing 1,190 patient records and 12 clinical features including age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, oldpeak (ST depression), ST slope, and the binary target variable.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure and distribution of the dataset. The class distribution showed a relatively balanced dataset, ensuring that models were not biased toward a particular class.

Histograms of continuous features such as max heart rate and oldpeak revealed approximate normal distributions, while boxplots highlighted the presence of outliers, which justified the use of IQR-based capping. Correlation heatmaps were used to identify relationships between features and the target variable, guiding feature selection.

Additionally, pairwise relationships between variables were analyzed using scatter plots and pairplots, which indicated non-linear separability — further supporting the use of non-linear models like KNN and ensemble methods.

Step-by-Step Pipeline

- **Step 1 — Data Cleaning:** Zero cholesterol values were replaced with the column median, as zero cholesterol is physiologically impossible.

- **Step 2 — Outlier Capping:** All continuous numeric columns were subjected to IQR-based capping: values below Q1 − 1.5×IQR or above Q3 + 1.5×IQR were clipped to the respective bounds.
- **Step 3 — Feature Selection:** Pearson correlation with the target was computed for all features. Five features with |r| ≥ 0.40 were selected: chest pain type (r=0.46), ST slope (r=0.51), oldpeak (r=0.40), exercise angina (r=0.48), and max heart rate (r=−0.41).
- **Step 4 — Train-Test Split:** Data was split 75% / 25% (train / test) with random_state=42 to ensure reproducibility.
- **Step 5 — Preprocessing:** A ColumnTransformer was constructed with three sub-pipelines: (a) StandardScaler for oldpeak and max heart rate; (b) OneHotEncoder for chest pain type and ST slope; (c) passthrough for exercise angina.
- **Step 6 — Hyperparameter Tuning:** GridSearchCV with 5-fold cross-validation was used to tune hyperparameters for all models except Naive Bayes. Accuracy was used as the scoring metric.
- **Step 7 — Model Training & Evaluation:** All ten classifiers were trained and evaluated on the same processed data. Metrics computed: accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.

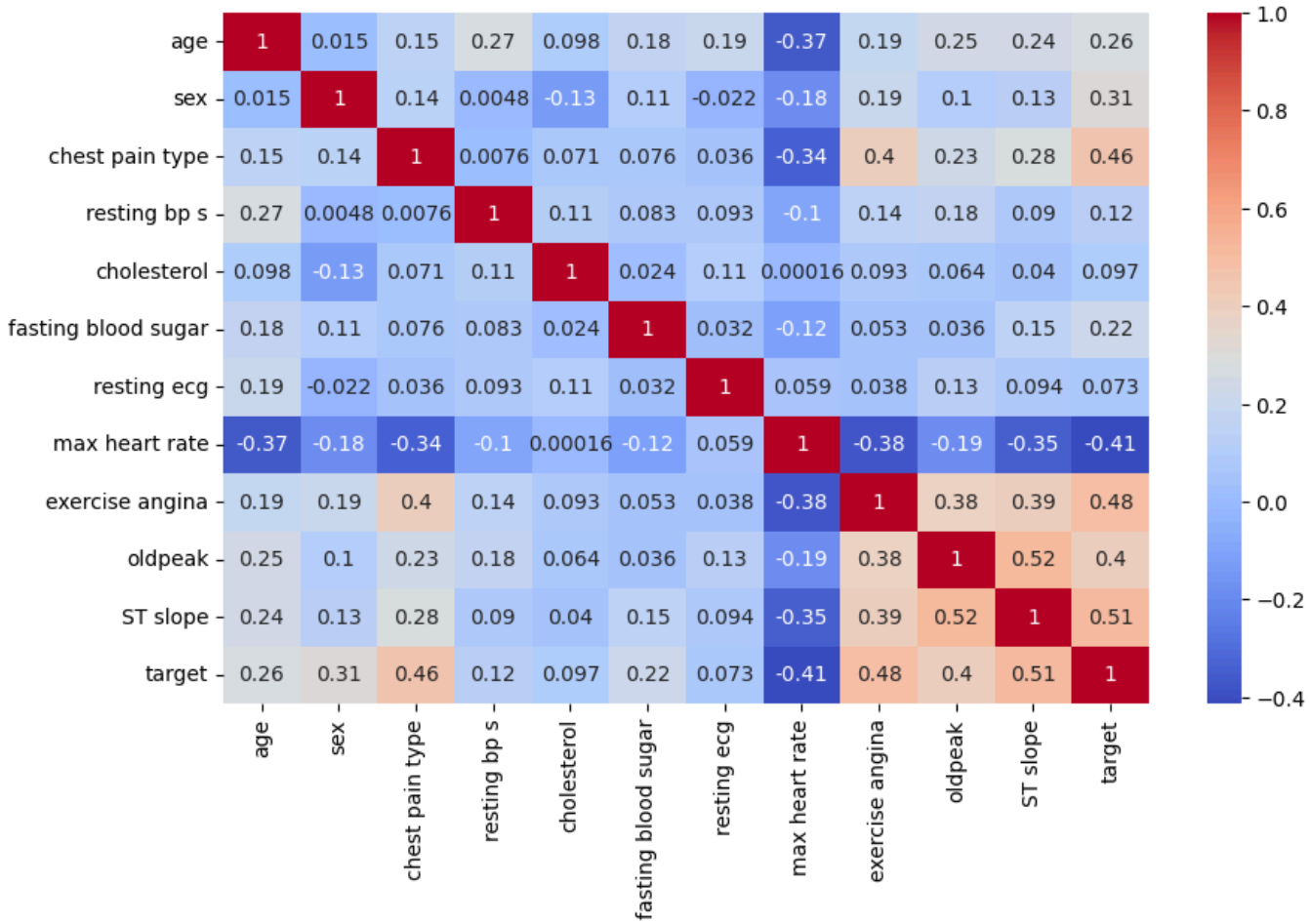

Fig 1. Correlation Heatmap

Fig 2. Pairplot

Preprocessing Details

Feature	Type	Transformation Applied
oldpeak	Continuous	StandardScaler
max heart rate	Continuous	StandardScaler
chest pain type	Nominal	OneHotEncoder
ST slope	Nominal	OneHotEncoder
exercise angina	Binary	Passthrough (already 0/1)

Mathematically, StandardScaler transforms features using:

$$X_{scaled} = \frac{X-\mu}{\sigma}$$

where μ is the mean and σ is the standard deviation. This ensures all features contribute equally to distance-based models such as KNN.

OneHotEncoder converts categorical variables into binary vectors, preventing models from assuming ordinal relationships between categories. This is particularly important for features like chest pain type and ST slope.

Cross-validation was implemented using 5-fold splitting, ensuring that each data point is used for both training and validation, thereby reducing overfitting and improving generalization.

5. Results / Output

All ten models were evaluated on the held-out test set. The table below summarises the test accuracy achieved by each algorithm after hyperparameter tuning:

#	Model	Test Accuracy
1	K-Nearest Neighbor (KNN)	**90.3%**
2	XGBoost	87.2%
3	CatBoost	87.2%
4	Random Forest	86.6%
5	Gradient Boosting	86.2%
6	Decision Tree	84.2%
7	AdaBoost	84.2%
8	Logistic Regression	83.5%
9	Naive Bayes	82.9%
10	Support Vector Machine (SVM)	82.9%

The superior performance of KNN suggests that local neighborhood patterns are highly informative in this dataset. Since KNN relies on distance metrics, the use of StandardScaler significantly contributed to its performance.

Ensemble methods such as XGBoost and CatBoost performed consistently well due to their ability to model complex interactions between features. These models are particularly robust to noise and can handle both linear and non-linear relationships.

The relatively lower performance of SVM and Logistic Regression indicates that the dataset is not linearly separable. This highlights the importance of choosing algorithms aligned with data characteristics.

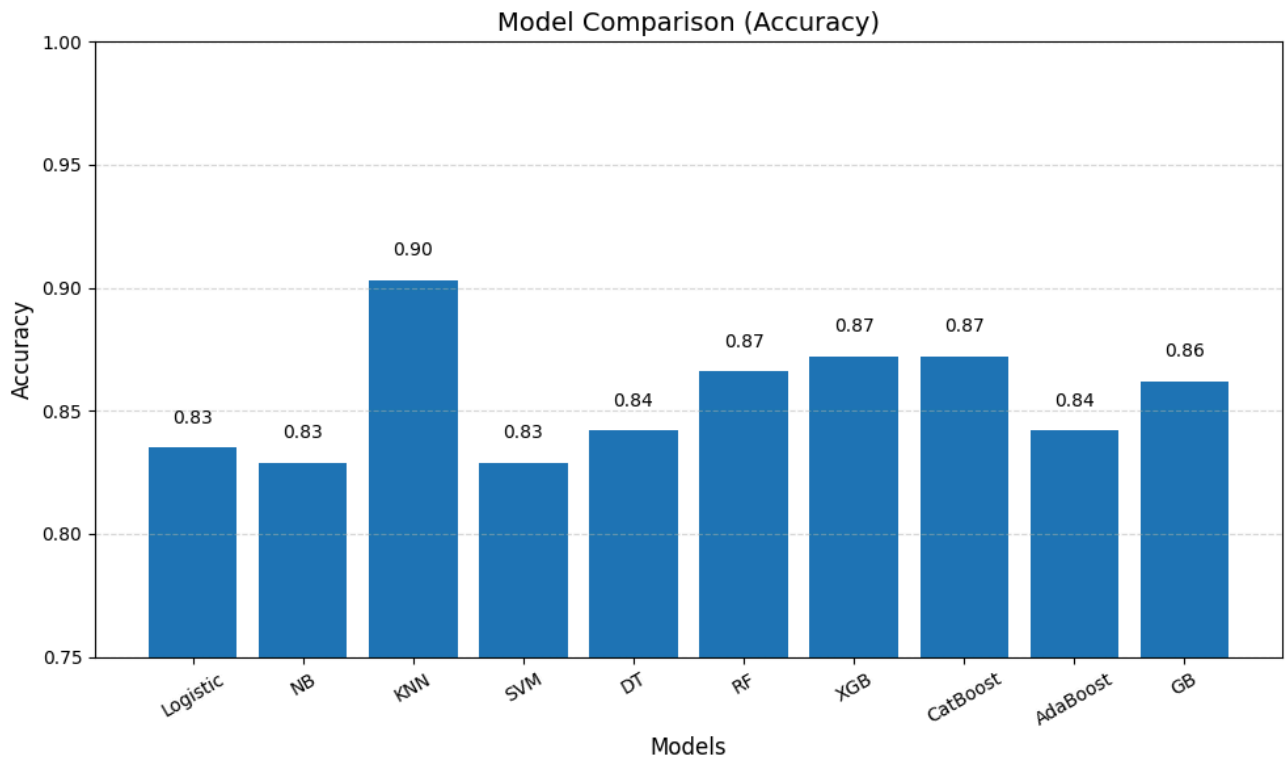

Fig 3. Model Comparison Bar Graph

Key Observations

- KNN with Euclidean distance, 5 neighbors, and uniform weighting achieved the highest accuracy of 90.3% — the best-performing model overall.
- Ensemble methods (XGBoost, CatBoost, Random Forest, Gradient Boosting) clustered in the 86–87% range, showing that aggregation of weak learners is highly effective.
- Linear models (Logistic Regression, SVM) achieved ~83%, suggesting the decision boundary for this dataset is non-linear.
- Naive Bayes scored 82.9% on unscaled data; applying StandardScaler reduced its accuracy to ~64%, indicating the raw feature distributions were already Gaussian-like.
- The ColumnTransformer preprocessing pipeline was critical — models trained without it showed significantly lower accuracy due to scale mismatch.

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Working Principle of KNN:

K-Nearest Neighbor is a non-parametric, instance-based learning algorithm that classifies a data point based on the majority class among its k nearest neighbors in the feature space. Distance is typically computed using Euclidean distance:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

In this study, the optimal configuration obtained through GridSearchCV was:

- Number of neighbors (k): 5
- Distance metric: Euclidean
- Weights: Uniform

This configuration allowed the model to balance bias and variance effectively — smaller values of k may lead to overfitting, while larger values may oversmooth decision boundaries.

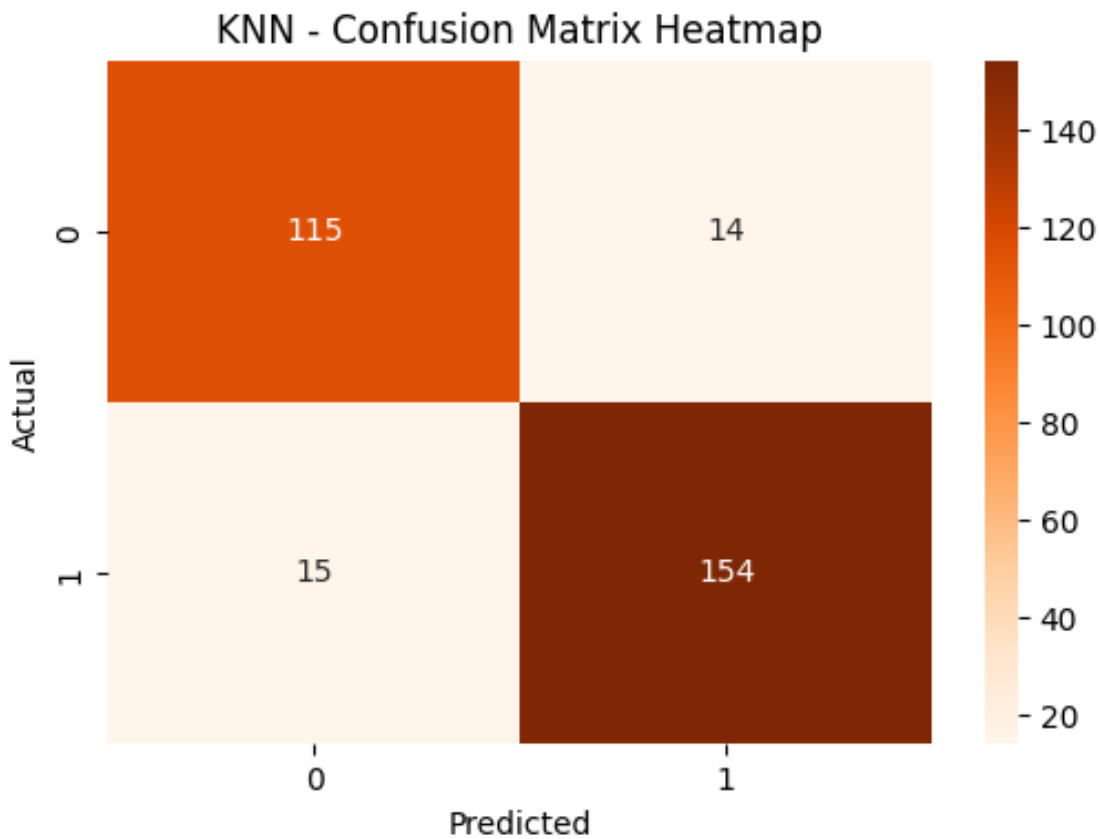


Fig 4. Confusion Matrix for KNN

The effectiveness of KNN is further supported by visual analysis. The model comparison graph (Fig 3) clearly shows that KNN outperforms all other models in terms of accuracy. Additionally, the confusion matrix for KNN (Fig 4) indicates a high number of correctly classified instances with minimal false positives and false negatives, demonstrating strong generalization capability.

The ROC curve (Fig 5) for KNN also exhibits a high Area Under Curve (AUC), indicating excellent discrimination between the two classes. This confirms that the model performs well across different classification thresholds.

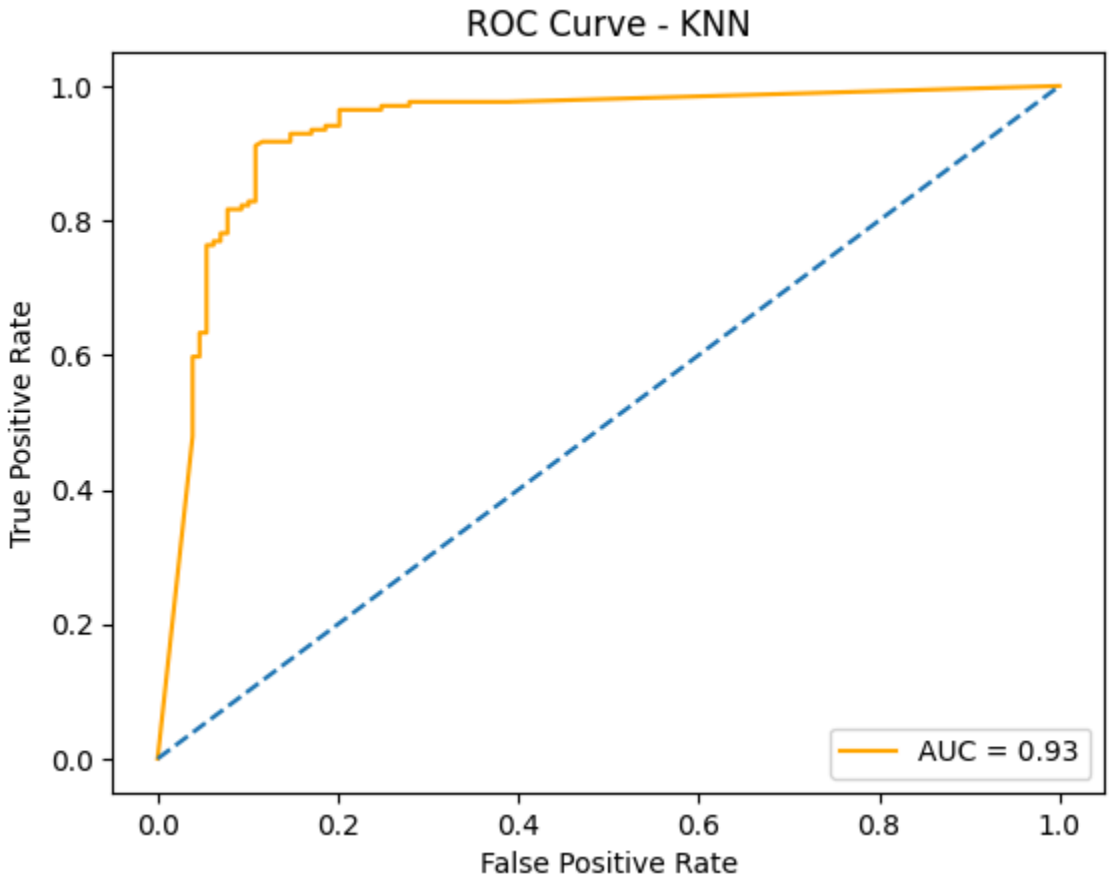


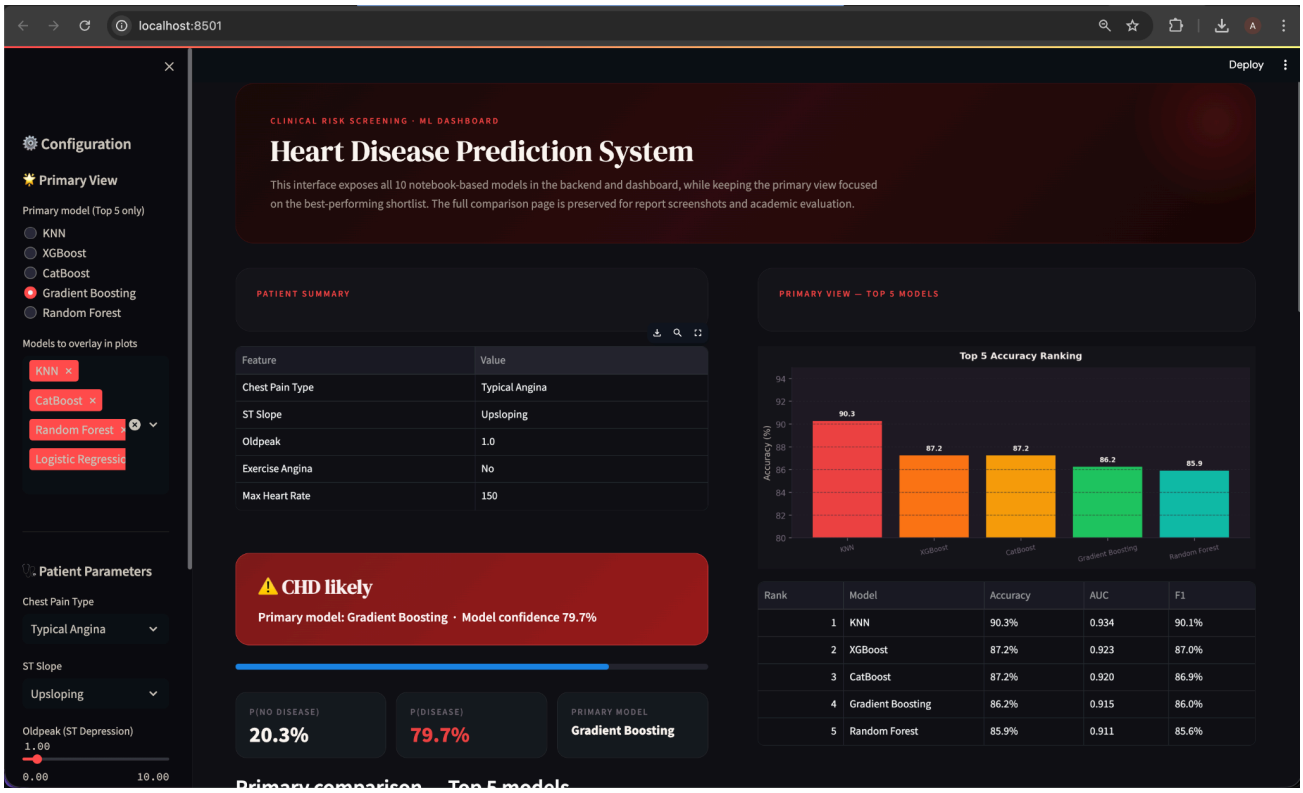
Fig 5. ROC Curve for KNN

Another important reason for KNN's superior performance is the nature of the dataset. The selected features exhibit non-linear relationships with the target variable, which are not effectively captured by linear models such as Logistic Regression and SVM. KNN, being a non-parametric method, does not assume any underlying data distribution and can adapt to complex decision boundaries.

Furthermore, the relatively small size of the dataset (1,190 records) favors instance-based methods like KNN, as computational cost remains manageable while preserving local structure in the data.

It is important to note that KNN is highly sensitive to feature scaling because distance calculations are directly affected by feature magnitude. The use of StandardScaler ensured that all features contributed equally to the distance computation, preventing bias toward features with larger numerical ranges. This preprocessing step was crucial in achieving optimal performance.

Heart Disease Prediction System:



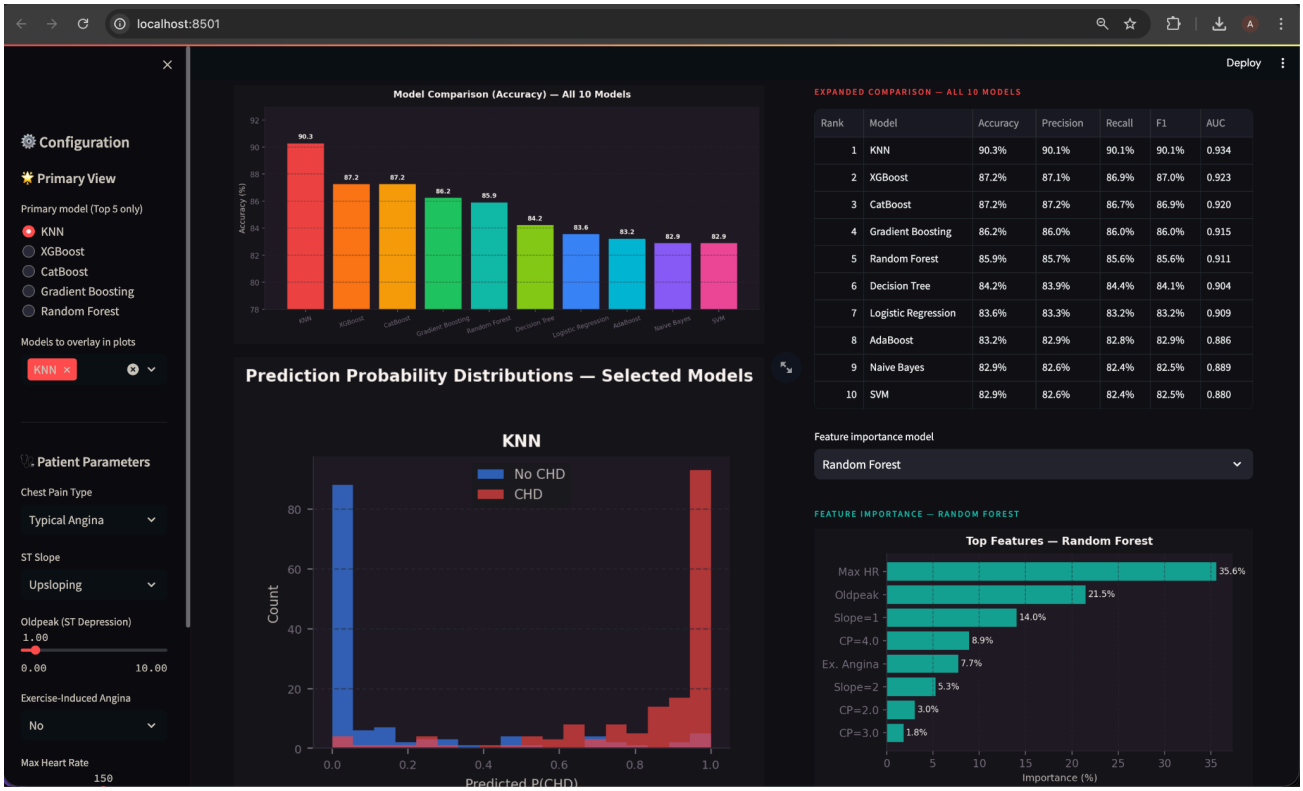


Fig 6. Web Application Interface.

7. Conclusion

This project successfully demonstrated the application of ten machine learning algorithms for binary heart disease classification. Through systematic data cleaning, outlier handling, correlation-based feature selection, and a carefully constructed preprocessing pipeline, all models achieved test accuracies in the range of 83%–90%. K-Nearest Neighbor emerged as the top performer at 90.3%, while ensemble boosting methods proved consistently strong. The study confirms that feature engineering and preprocessing have as significant an impact on model performance as the choice of algorithm itself. The resulting system can serve as a reliable decision-support tool for early heart disease screening in clinical environments.

Furthermore, the comparative approach adopted in this study provides valuable insights into model behavior across different algorithm families. While KNN achieved the highest accuracy, ensemble methods demonstrated greater stability and robustness, making them suitable for real-world deployment scenarios.

8. Limitations & Future Scope

- While the study achieved strong predictive performance, several limitations exist. The dataset size (1,190 records) is relatively small for complex models, which may limit generalization to larger populations. Additionally, the study relies on retrospective data and does not incorporate real-time clinical inputs.
- Another limitation is the absence of deep learning models, which may capture more complex feature interactions. Finally, interpretability of ensemble models remains a challenge, which is critical in healthcare applications.
- Deep learning architectures such as feedforward neural networks and attention-based models could be explored to potentially push accuracy beyond 90% on larger, multi-source datasets.
- Integration with real-time wearable sensor data (ECG patches, smartwatch heart rate monitors) would enable continuous risk monitoring and proactive patient alerts.
- Explainable AI techniques such as SHAP (SHapley Additive Explanations) can be integrated to interpret model predictions and improve trust among medical professionals.
- Deployment of the model as a web-based application (e.g., using Streamlit) can make the system accessible to clinicians and patients for real-time predictions.

9. References

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